Student information Sender

Course Code – SE-409

Answer to the question no – 2 (a)

In my project I used Java Collection to store and manage data efficiently. The main collection I used was hashmap and sometime Array / ArrayList for storing multiple records.

→ why Hashmap was choosen?

⇒ I used → HashMap<Integer, Student>

because →

i) It stores data in key value pairs

ii) Student Id can be used as a unique key.

iii) Searching data become very fast.

iv) It helps to find duplicate student ID.

Example →

HashMap <Integer, Student> students = new HashMap<>();

Here →

Key → student ID

Value → student object

→ Why Array or Arraylist Chosen?

⇒ Array or arraylist used to:

Store Mutiple student record sequentially.

Display all result easily.

Example →

ArrayList<Student> list = new Arraylist<>();

→ How collection helped ??

⇒ Java collection help the program to →

- Add student information dynamically.
- Search student records quickly.
- Update or delete data easily.
- Organize result efficiently.

So, using collection make the program more flexible and easier to maintain.

Ans to the question no- 2(b)

Socket communication was used to establish communication between the client and the server. The server waits for connection request, and the client connects to the server using an IP address and port number. After connection both sides can send and recive data.

→ Roles of server and Server Socket.

Server socket →

Used to on the server side.

→ Listen for incomming client request.
→ Create a communication chanel.
→ Accept client connection.

Example →

```
ServerSocket serverSocket =
    new ServerSocket(5000);
```

Socket →

Socket is used on both client and server side.

→ For sending data.

→ For receiving data.

→ Maintain communication between client and server.

Example →

```
Socket socket = new Socket("localhost", 5000);
```

Answer to the question no – 2©

Steps to send and receive data using socket →

Step: 01: → Create ServerSocket on Server.

The ~~Sever~~ server create server socket using a port number.

Ex→ ServerS.dat server = new ServerSocket (5000);

Step: 2 → Accept client connection.

The server waits for the client connection.

Ex →

Socket socket = server.accept ();

Step:03 → Create socket on client.

The client connects to the server using ip address and port number.

Step:04 → Create input and output stream.

Ex →

```
DataInputStream dis = new
    DataInputStream(Socket.getInputStrea
                    m());

DataOutputStream dos = new DataOutput
    Stream(Socket.getoutStream());
```

Step:05 → Send Data

The client sends data using output Stream.

Example →

```
dos.writeUTF("Hello Server");
```

Step: 6 → Recive data

The receiver reads data using input stream.

Example →

```
String msg = dis.readUTF();
```

Step-07 → Close Connection.

Conclusion → Java collection helped (manage and organize student data efficiently. while socket programming enabled communication between client and server.